1. ***This is the start of a multipart question. The information below pertains to question 1 to question 4.***

   NVDA is the ticker symbol for NVIDIA. Consider the weekly returns for this stock between March 31, 2019 and December 07, 2024.

   The following code creates a vector `x` containing the weekly returns:

```
library(quantmod)
library(dplyr)

x = getSymbols("NVDA",
               from = "2019-03-31",
               to = "2024-12-07",
               auto.assign = FALSE) %>%
  Ad() %>%
  weeklyReturn() %>%
  as.data.frame() %>%
  pull(weekly.returns)
```

   The returns are calculated using the adjusted closing price, which accounts for dividends and stock splits. `x` should contain 297 weekly returns.

   What was the minimum weekly return over this time period?

   a. -12.96%
   b. -8.30%
   c. -12.28%
   d. -16.07%
   e. None of the above

2. Given the information above.

   What was the interquartile range of weekly returns over this time period?

   a. 1.64%
   b. 3.72%
   c. 7.96%
   d. 11.85%
   e. None of the above

3. Given the information above.

   What is the 0.05-quantile of the weekly returns?

   a. -14.60%
   b. -8.59%
   c. -6.63%
   d. -5.46%
   e. None of the above

4. Given the information above.

   What is the empirical probability that the weekly return is less than zero?

   a. 0.6301
   b. 0.4343
   c. 0.5470
   d. 0.6191
   e. None of the above

5. ***This is the start of a multipart question. The information below pertains to question 5 to question 6.***

   PEP is the ticker symbol for PepsiCo. Consider the monthly returns for this stock between January 01, 2013 and June 30, 2026.

   The following code creates a vector `x` containing the monthly returns:

```
library(quantmod)
library(dplyr)

x = getSymbols("PEP",
               from = "2013-01-01",
               to = "2026-06-30",
               auto.assign = FALSE) %>%
  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  pull(monthly.returns)
```

   The returns are calculated using the adjusted closing price, which accounts for dividends and stock splits. `x` should contain 162 monthly returns.

   We would like to compare the empirical distribution of monthly returns to a normal approximation with mean 0.008 and standard deviation 0.043.

   Focus on the 0.58-quantile. This is the return value such that approximately 58% of monthly returns are less than or equal to it. Which pair correctly reports the empirical 0.58-quantile and the 0.58-quantile from the normal approximation?

   a. The empirical 0.58-quantile is 2.31%, and the normal approximation 0.58-quantile is 1.57%
   b. The empirical 0.58-quantile is 2.77%, and the normal approximation 0.58-quantile is 1.40%
   c. The empirical 0.58-quantile is 1.95%, and the normal approximation 0.58-quantile is 1.67%
   d. The empirical 0.58-quantile is 2.77%, and the normal approximation 0.58-quantile is 1.57%
   e. None of the above

6. Given the information above.

   The CDF is defined as $F(a) = \Pr(x \leq a)$. In this context, $F(a)$ is the probability that the monthly return is less than or equal to a. Evaluate the CDF at $a = 0.03$. Which pair correctly reports the empirical probability and the probability implied by the normal approximation?

   a. The empirical probability is 0.9961, and the normal approximation probability is 0.6751
   b. The empirical probability is 1.2180, and the normal approximation probability is 0.5102
   c. The empirical probability is 0.7099, and the normal approximation probability is 0.5102
   d. The empirical probability is 0.7099, and the normal approximation probability is 0.6955
   e. None of the above